MILITARY SPECIFICATION

MANUAL SET, TECHNICAL, ORGANIZATIONAL AND INTERMEDIATE MAINTENANCE; GENERAL REQUIREMENTS FOR PREPARATION OF (FOR M-X)

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers the requirements for the preparation of technical manuals that comprise the organizational and intermediate level manual sets for the Missile X (M-X) weapon system.

1.1.1 Organizational and intermediate maintenance manual sets. Collectively, the manuals shall contain all essential information required to maintain and support the systems and equipment for which these manuals are prepared. Each manual set shall contain all instructions for the work involved recognizing that some of the data may be duplicated in other manuals specified herein. This is required so that each system specialist, for example, has in one manual set, instructions for all required work on the system. While the scope of this specification specifically addresses organizational and intermediate level maintenance activities, it may also be used to include operation of certain M-X unique systems or equipment. These systems include such equipment as operational support vehicles. Information required to operate such systems or equipment shall be included in the appropriate manuals of the organizational level maintenance manual sets in accordance with the general formats described herein. The organizational and intermediate maintenance manual sets are comprised of those manuals necessary to support system maintenance. These manuals are identified and defined in Section 6.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Ballistic Missile Office/AWD, Norton AFB, CA 92409 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Military

MIL-M-38784	Manuals, Technical: General Style and Format Requirements
MIL-P-38790	Printing Production of Technical Manuals: General Requirements for

HANDBOOKS

Military

MIL-HDBK-275	Guide for Selection of Lubricants, Fluids, and Compounds for use in Flight Vehicles and Components

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

Handbook H4-1	Federal Supply Code for Manufacturing (United States and Canada) Name to Code
Handbook H4-2	Federal Supply Code for Manufacturing (United States and Canada) Code to Name

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General.

3.1.1 Format. The general format requirements detailed in this specification and the general requirements of MIL-M-38784, are applicable.

3.1.2 Style of writing. The writing style requirements of MIL-M-38784, are applicable.

3.1.3 Nomenclature. When full or formal system nomenclatures are used, they shall be followed by the applicable higher level reference designation in parentheses. These designators need not be repeated on a text page. Chapter/section titles and

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system/subsystem nomenclatures shall be followed by the applicable system/subsystem/subject number in parentheses. If required, a cross reference may be added to list the official nomenclature and its corresponding acronym or general usage nomenclature.

3.1.4 Front matter. The front matter shall be as specified in MIL-M-38784 as modified by Figures 1, 2, 3, 4 and 5, or as specified by the procuring activity.

3.1.5 Illustrations. The illustration requirements of MIL-M-38784, are applicable. All illustrations shall be line art. Halftone illustrations shall not be used.

3.1.6 Diagrams. System wiring data and system schematics shall be developed in accordance with MIL-M-38784.

3.1.7 Foldout pages. The page number, system/subsystem/subsubsystem number (S/S/SN), figure number and figure title for foldout pages shall be visible when the printed page is folded. Page sizes shall conform to the requirements described in MIL-M-38784.

3.1.8 Marginal copy. The applicable S/S/SN shall appear in the lower, outer corner of each page, directly above the page number and shall be approximately 18 point type (Figures 6, 7 and 8). S/S/SN shall not be placed at the bottom of front matter pages, i.e., title, A page, table of contents.

3.1.9 Security classificaton markings. The security classification markings shall be in accordance with MIL-M-38784.

3.1.10 Nuclear hardness and survivability (NH&S). NH&S markings will be in accordance with MIL-M-38784.

3.1.11 Classified material. Every effort shall be made to include only unclassified information in these manuals. If classified information is essential, separate classified supplemental manuals will be prepared in accordance with MIL-M-38784.

3.1.12 Operating procedures for test equipment. Instructions essential for proper use of test equipment shall be included within each of the procedures: however, standard procedures for operating oscilloscopes, voltmeters, multimeters, etc., shall not be included.

3.2 General weapon (GW) manual. Multiple GW manuals shall be prepared for the M-X weapon system. Manuals may be structured in sets for user convenience. For definition, see 6.3.1.

3.2.1 Scope. The GW manual shall provide general information to be used by maintenance personnel to further understand the system. Specifically, it shall provide:

a. An overview and detailed explanation of the organizational and intermediate maintenance manual set

b. Functional description of the S/S/SN system

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c. An overall description (applicability of model and system serial number), general arrangement, principal dimensions, station diagrams, access doors, danger areas

d. A summary description of all general systems in the weapon system supported by appropriate artwork

e. Non-Job Guide (JG) Manual information applicable to more than one system (3.2.4.2.3).

f. Non-JG formatted information and instructions that are general to the system

g. General information on interfaces between systems and integration of two or more systems. Block diagrams of the system interfaces shall be provided.

3.2.2 Numbering. Sections, paragraphs, pages, figures, and tables shall be numbered in accordance with MIL-M-38784.

3.2.3 Page size. The GW manual shall be prepared on standard 8-1/2 by 11 inch page size paper.

3.2.4 Arrangement. The basic content and arrangement of the GW manual shall be as follows:

a. Front Matter (See Table I)

b. Introduction

c. Section 1 - General Weapon Description

d. Section 2 - Organizational and Intermediate Level Maintenance Manual Set

e. Section 3 - (and subsequent sections) Address such topics as: dimensions and areas, facility or shelter descriptions, deployment area description, etc.

3.2.4.1 Introduction. The introduction shall provide a brief explanation of the manual's purpose, scope, and arrangement. Include any relevant information that increases the usability of the manual. The use of "shall", "will", "should", and "may", and also the definitions of "warnings", "cautions", and "notes", as defined in MIL-M-38784, shall be explained in the introduction.

3.2.4.2 Sections

3.2.4.2.1 Section 1. Section 1 shall provide an overall description of the M-X weapon system and its general subsystems. The general arrangement, principal dimensions, station diagrams, walkways, access doors, roadways, and danger areas shall be described. Adequate illustrations including first level system block diagrams (selected from the Schematic Diagram (SD) Manual if appropriate) will be used to aid understanding and enable the manual to stand independently.

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Table I. Front Matter

Front Matter Description	General Weapon (GM)	General System (GS)	Sequence Chart (SC)	Job Guide (JG)	Job Guide Index (JX)	Fault Tracking (FT)	Fault Isolation (FI)	Wiring Data (WD)	Schematic Diagram (SD)	Schematic Diagram Index (SX)	Wiring Data Index (WX)
Title Page Figs 1, 3, 4, and 5	•	•	•	•	•	•	•	•	•	•	•
List of Effective Pages ("A" page)	•	•	•	•	•	•	•	•	•	•	•
TO Verification Requirements page ("b" page)		•	•	•							
TO/Equipment Configuration Status Record ("c" page)		•	•	•		•	•	•	•		
Table of Contents	•	•	•	•		•	•	•	•	•	•
List of Illustrations	•	•	•	•			•	•	•		
List of Tables	•	•		•	•	•	•	•			
Alphabetic Index	•	•				•	•	•		•	•
Warning/Safety Summary Page (Fig 2)		•		•			•				

3.2.4.2.2 Section 2. This section shall contain a list by technical order (TO) number and title of the complete set of manuals that make up the organizational and intermediate maintenance manual set. Section 2 shall also contain an overview and detailed explanation of related maintenance manuals, the use of TO numbering, and assignment of higher reference designators that relate to a functional system, definition of a functional system, the TO integration methodology, and detailed

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explanation and intended use of each type of manual making up the maintenance manual sets. Illustrations will be used to assist in explanation as required.

3.2.4.2.3 Section 3 and subsequent sections. These sections shall provide information that is not directive in nature and does not facilitate job guide formatting. These sections shall include information applicable to more than one system, system interfaces, integration of two or more systems, etc. Block diagrams of the system interfaces shall be provided.

3.3 General system (GS) manual. Separate GS manuals shall normally be prepared for each general system. Coverage of a specific subsystem shall not be divided between two GS manuals. For definition, see 6.3.2.

3.3.1 Scope. The GS manual shall provide system, subsystem, and subsubsystem description, theory of operation, and maintenance support information. The GS manual shall include purpose, type, main features, and supporting data for the system being covered. The level of detail of information shall be consistent with the approved organizational and intermediate level maintenance concepts.

3.3.2 Numbering. Sections, paragraphs, pages, figures and tables shall be numbered in accordance with MIL-M-38784. Each manual will be divided by sections as defined in 3.3.4.

3.3.3 Page size. The GS manual shall be prepared on standard 8½ by 11 inch page size paper.

3.3.4 Arrangement. The GS manual shall be arranged in the following manner:

a. Front Matter (Table I)

b. Introduction

c. Section 1 - System Functional Description organized by S/S/SN reference number. (System functional description might require more than one section for proper coverage. In this case the following sections move to consecutive higher section numbers.)

d. Section 2 - General Maintenance Requirements.

e. Section 3 and subsequent sections as required:

f. Subsystem/subsubsystem description and theories of operation.

g. Last Section - Support Equipment, Special Tools and Supplies.

3.3.4.1 Introduction. The introduction shall be arranged in the following manner:

a. Purpose of Manual

b. Scope of Manual

c. Arrangement of Manual

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d. Use of Manual

e. List of Abbreviations and Definition of Terms

f. Missile or Other System Effectivity Codes

3.3.4.2 Sections. The GS manual may be sectionalized by system, subsystem, or subject number depending on degree of complexity. The last section of the GS manual shall be reserved for two tables: (1) support equipment and special tools, (2) supplies (consumables), and for fabrication information for local manufacture of support equipment and special tools.

3.3.4.2.1 Section 1. Section 1 shall describe the theory of operation for the system. Section 1 shall be based on functional flows within the system, and shall be supported by simplified schematics. Block diagram or simplified schematics shall be used as necessary to support the system description. Section 1 shall contain illustrations showing the appropriate locations of all line replaceable units (LRUs). For complex systems where components of the system control or provide inputs to or otherwise interface with various subsystems/subsubsystems, Section 1 shall describe the complete system and the components that make up the various subsystems/subsubsystems.

3.3.4.2.2 Section 2. Section 2 shall provide general maintenance information that is applicable to the general subsystem but is not suitable for JG format. Section 2 shall include general information that is nondirective in nature consisting of such information as: safety wiring procedures, bonding, torque wrench application, grounding, fasteners, adhesive, sealants, hydraulic or pneumatic system fitting definitions with illustrations, connector application, etc. Specific or peculiar instructions for such items as leakage rates, torque application, safety wiring, cleaning, etc., will be contained in the appropriate JG manual.

3.3.4.2.3 Section 3 and subsequent sections. These sections shall provide a description and the theory of operation for the subsystem/subsubsystem being covered. The theory of operation may be combined for several subsystems within a system. Appropriate references shall be made to that section. For noncomplex systems, each section shall provide a description and the theory of operation for the S/S/SN being covered. Each section shall contain illustrations necessary to support system descriptions.

3.3.4.2.4 Last section. This section of the GS manual shall include a support equipment and special tools list, local manufacture instructions and a supplies list. The introduction to the section will include the following statement:

"Alternate Equipment. Equivalent Air Force standard test equipment may be used:

a. Except when specifically designated test equipment is called for.

b. If identified in the USAF Interchangeability and Substitution (I&S) grouping stock list, and not identified in this manual.

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c. If authorized by Precision Measurement Equipment Laboratory (PMEL), Technical Engineering (TE) or the AFLC system manager (SM) as a suitable substitute but not listed in USAF I&S.

When PMEL or TE approves a substitute, the requesting unit will forward this information to the AFLC SM and HQ SAC/LGBM.

The same authority may be exercised by TE for all other equipment and special tools. Extreme care shall be used in selection and use of substitute (equivalent) equipment to insure that system or equipment degradation does not result from the use of the substitute."

3.3.4.2.4.1 Support equipment and special tools. A consolidated list of government approved support equipment and a consolidated list of special tools specified in the JG manual set shall be provided. Illustrations shall be provided for system-unique test equipment as required. Common tools (screwdrivers, pliers, side cutters, soldering irons, etc.) shall not be listed. Common support equipment (voltmeters, signal generators, maintenance platforms, etc.) shall be listed. The list shall be in the format described below:

Part No. (Mfr. Code)	Nomenclature	Alternate (Mfr. Code)	Application

a. Part No. (Mfr. Code). List the part number when available. The National Stock Number (NSN), may be listed when the part number is not available. The Manufacturer's Code, as listed in HDBK-H4-1 and -2, shall follow the part number.

b. Nomenclature. Nomenclature shall agree with the drawing title (but shall be forward reading) and shall be consistent in all related manuals. Type designators shall be included as applicable.

c. Alternate. When no equivalent exists and there is no acceptable alternate, enter "None." When only one acceptable alternate exists, list it. When several alternates exist, but one (or more) is preferred, list the preferred alternate and also add "Equivalent." When several alternates exist and none are preferred, enter "Equivalent."

d. Application. The application (use) of the support equipment or tool and the appropriate manual shall be listed.

3.3.4.2.4.2 Local manufacture instructions. An illustration shall be provided for each item of local manufacture. Each illustration shall provide complete information for fabricating, assembling, and acceptance testing for each item of locally manufactured support equipment. Each illustration shall contain a list of materials and notes sufficient to manufacture the part or assembly. The list of materials shall include: quantity required per unit, locator number (callout number) to find the item on the drawing, nomenclature, local manufacturer code, part number (identifying number or military standard), specification number.

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3.3.4.2.4.3 Supplies (consumables). A consolidated list shall contain all consumables specified in the organizational maintenance JG set for the general system being covered. The list shall be arranged alphabetically in the following format:

Nomenclature	Specification/ Gov't Std No.	Part No. (Mfr. Code)	Application

a. Nomenclature. List the generic nomenclature, such as: lubricant, gasket, sealant, etc.

b. Specification/Gov't Std No. List the military or commercial specification or government standard as applicable.

c. Part No. (Mfr. Code). If specification or government standard is not available, list the manufacturer's part number and code from Handbook H4-1 or -2. If manufacturer's code is unassigned, list manufacturer's name and address.

d. Application. List the application (use) of the consumable and the appropriate manual.

3.4 Sequence chart (SC) manual. SC manuals shall be prepared for the weapon system. The SC manual may be structured in sets for user convenience. For definition, see 6.3.3.

3.4 1 Scope. The SC manual shall provide diagrams to guide maintenance personnel in placing the equipment/system in the proper mode for the maintenance required, then returning the equipment/system to the desired mode.

3.4.2 Numbering. Sections, paragraphs, pages and figures shall be numbered in accordance with MIL-M-38784. Each manual will be divided by sections as defined in 3.4.4.

3.4.3 Page size. The SC manual shall be prepared in standard 8-1/2 by 11-inch page size format using standard size foldouts.

3.4.4 Arrangement. The SC manual shall be arranged in the following manner:

a. Front Matter (Table I)

b. Introduction

c. Section 1 - Those sequence charts required for scheduled maintenance and similar activities shall be included in Section 1.

d. Section 2 - Those sequence requirements required for unscheduled maintenance activities shall be included in Section 2.

3.4.5 Format. Sequence charts may be single page or standard foldout page format. Maintenance job guide functions shall be entered by TO number and function number. Refer to Figure 9 for further format guidelines.

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3.5 Job guide (JG) manual. The JG manuals shall provide complete, detailed procedures for organizational and intermediate maintenance. These procedures shall be organized into a series of manuals. A separate manual shall be prepared for each general system. Separate manuals may also be prepared for subsystems depending on the complexity of the equipment. JG manuals shall comply with the style and format defined and illustrated herein. For definition, see 6.3.4.

3.5.1 Scope. The JG manual shall provide complete start-to-finish, step-by-step detailed maintenance procedures arranged in logical sequence of occurrence. These instructions will be adequately illustrated to assure the intended user's understanding of what is to be accomplished. Typical instructions covered in JGs are: removal, handling, repair, installation, adjustment, calibration, operational checkout and other maintenance procedures as applicable plus operation of special equipment addressed in paragraph 1.1.1.

3.5.2 General requirements for JG manuals.

3.5.2.1 Level and style of writing. The information in each JG manual shall be sufficiently complete, clearly worded and adequately illustrated so maintenance personnel, inexperienced on the particular equipment, but with general training and experience on similar equipment, are able to maintain the equipment efficiently. Emphasis shall be placed on specific steps to be followed. Reading grade levels shall be as specified by MIL-M-38784. Second person imperative mood shall be used for maintenance instructions, for example: set POWER switch to ON. Third person indicative mood or second person imperative mood may be used for descriptions or discussions and for notes, cautions and, where appropriate, warnings. Articles (such as: an, a, the) should be excluded.

3.5.2.2 Syntax. Normal sentence structure (subject, verb, object) shall be used for explanations, discussions and statements. Directive steps will be action (verb) oriented and shall be structured as follows: (1) subject, (2) verb, (3) object, (4) and (5) adverb or adverbial phases as necessary.

a. Subject. Implicit, except when more than one technician is required.

b. Verb. Select a verb that best describes the action required by the technician, or that best describes the technicians behavior with respect to the object.

c. Object. The specific equipment to which the technician's action is directed.

d. Adverb/adverbial phrase. Used as necessary to define: (1) location of equipment to be manipulated or (2) method or direction of manipulation.

(b) (c) (d1) (d2)

Example: Set NORM switch on IFF console to OFF.

3.5.2.3 Nomenclature. Nomenclature consistency shall be maintained throughout the JG manuals related to a system. Nomenclature used in the text shall correspond with that appearing on the equipment decals, engraved legends, nameplates, etc. When nomenclature does not appear on the equipment, the nomenclature used on the

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engineering drawing shall be used in the manual. Nomenclature in the JG and its related GS manual shall be consistent.

3 5.2.4 Tolerances. Information on how the equipment must respond, either in numerical or quantitative terms will be provided as part of the text. All torque values, clearances, flow rates, pressures, voltages, etc., will be provided at each appropriate text location. Measurements shall be given in the units of the tool(s) or equipment required, i.e.; "Torque mounting bolts 50 (±10) in/lb and lockwire," "Check voltmeter for 120.0 (±2.4) Vac." Do not use percentage. General instructions for installation of standard hardware that does not require specifically measured values may be used provided complete instructions are given in the step. For example: "Install 40 panel screws and tighten so screw heads are flush with the panel and panel is flush with surrounding surface."

3.5.2.5 Page size and limitations. Final JG manuals shall be prepared in 8-1/2 by 11-inch size, and will be limited to approximately 400 pages (200 sheets). If the proposed JG exceeds the page limitations the procuring activity shall be requested to provide breakout instructions.

3.5.2.6 Foldouts. Foldout pages, when used in JG manuals, shall be authorized by written approval of procuring activity. Foldout illustrations, if used, will not exceed the maximum foldout size of 11 by 17 inches (Figure 10).

3.5.2.6.1 Blank aprons. Foldouts used in JG manuals need not have blank aprons when they support a single left-hand page of text (complete task). These foldouts will be right-hand pages located opposite the task instruction. Illustrations supporting a multi-page task shall have blank aprons and be located on a right hand page following the last applicable text page (Figure 10).

3.5.2.6.2 Text backed foldouts. Foldouts used in JG manuals shall not be backed with text.

3.5.3 Arrangement. The basic contents and arrangement of JG manuals shall be as follows:

a. Front Matter (Table I)

b. Introduction (when applicable)

c. Section 1 - (System peculiar general maintenance instructions)

d. Section 2 and subsequent sections (maintenance instructions)

3.5.3.1 Table of contents. The table of contents for JG manuals shall list each Line Replaceable Unit (LRU) or item covered, and the exact function title and task for that LRU or item. Each item shall be cross referenced to the S/S/SN index. The first manual of a multi-manual set shall list and cross reference the functions and tasks of all JG manuals in that set.

3.5.3.2 Introduction. A narrative introduction is required. The introduction shall contain information pertaining to the scope of the data included, abbreviations, and definition of terms used and effectivity codes. The statement regarding "Alternate

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Equipment" contained in 3.3.4.2.4 will be included in the JG introduction. In multi-manual JG manuals, the introduction shall be included in the first manual only.

3.5.4 Sections. Sections will be arranged as follows:

3.5.4.1 Section 1. When required, general maintenance procedures shall be in Section 1 of the JG manual. These procedures are peculiar to that system being covered, e.g., specific torque wrench usage, lockwire procedures, O-ring seal installation, leakage rates, equipment access and close-up functions. Procedures may be tabular or text format, supported by illustrations as required.

3.5.4.2 Section 2 and subsequent sections. Each section shall normally cover maintenance activities applicable to the subsystem being covered in the JG manual. A section may also cover an overall maintenance procedure such as system checkout.

3.5.4.2.1 Numbering. Sections shall be numbered consecutively within a JG manual and pages shall be numbered consecutively within each section. Each section may contain one or more functions.

3.5.5 Functions. A function consists of a task or a group of related tasks usually performed in the same sequence. The tasks are inseparable in any normal use of the function. Each maintenance function shall begin on a right-hand page followed by one or more tasks, then by follow-on maintenance instructions or reference to the sequence chart manual. The function title specifies the activity (remove, install, align, transport, etc.), identifies the system, subsystem or assembly as appropriate, and lists the reference designator, if applicable. For example:

Install Missile in Canister (S/S/SN)

Install Canisterized Missile on Transporter (S/S/SN)

3.5.5.1 Input conditions. This entry (Figure 6) shall include all necessary information needed to perform the function, as follows:

a. Applicability. This information shall specify that system or subsystem effectivities applicable to the function. The word "All" is used when the function applies to all systems or subsystems covered. An effectivity symbol may be used if explained clearly. Symbols shall be consistent throughout the JG series.

b. Required conditions. This paragraph identifies those functions or tasks that must be performed prior to the function being performed (if only a few functions are required). If a more complex series of activities must be performed, a sequence chart may be prepared and included in the SC manual. Reference shall then be made to the SC manual. The tasks or steps required to achieve a prerequisite conditon shall be consolidated into the procedural steps in lieu of being referenced when (1) the prerequisite procedure is less than a complete task; (2) the prerequisite condition is applicable to only a portion of the tasks comprising the function.

c. Support equipment and special tools. The Government approved support equipment (test equipment, special tools and ground handling equipment) applicable

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to the required maintenance shall be listed. Common types of tools normally found in the technician's tool kit, such as screwdrivers, pliers, side cutters, soldering irons, etc., shall not be listed. Common types of test equipment such as voltmeters, signal generators, etc., shall be listed. Support equipment will be listed by nomenclature or if not assigned, by the commercial or manufacturer's designation:

Part No. (Mfr. Code)	Nomenclature	Alternate (Mfr. Code)	Application

d. Supplies (consumables). Expendable items and materials required to accomplish the function shall be listed. Sealants, lubricants, replacement lamps, gaskets, seals, etc., are considered consumables. Attaching hardware requiring mandatory replacement shall be included. The list will consist of the items/quantities called out in all the JG tasks applicable to that function in tabular form as follows:

Nomenclature	Specification/ Gov't STD No.	Part No. (Mfr. Code)	Application	Quantity

Note: Consumables such as cheese cloth, plastic bags and masking tape may be identified by "COML" in the Part No. column.

e. Safety conditions. Information pertinent to safety shall be included and presented as a caution or warning. Activities requiring more than one person shall be identified with an appropriate warning or caution either preceding the step or in the introduction. Cautions and warnings pertaining to a complete function shall be provided as part of the input conditions. Cautions or warnings applicable to an individual task or step shall immediately precede the task or step to which they apply. Each warning or caution shall indicate the possible consequence for noncompliance, such as:

WARNING

Ensure battery cell covers are removed when charging battery to avoid explosion. Battery acid can severely burn skin or cause blindness. Remove battery cell covers.

3.5.6 Format. The JG introduction information is followed by the first task. Tasks shall consist of the task title followed by procedural steps with the necessary supporting illustrations. (Figures 7 and 8). Illustrations shall be placed in the outer column of the page. Text shall be placed in the inner column of the page. Illustrations shall be placed as close as possible to its associated text. When appropriate, a legend shall be provided with the illustration.

3.5.6.1 Numbering. Sections shall be numbered within the JG manual to cover a specific subsystem activity. Functions shall be numbered sequentially within a

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section. Tasks shall be numbered sequentially within functions. Each numerical identifier will be separated by a dash; i.e., the tenth function in the fourth section would be 4-10. The fourth task in the second function of section three would be 3-2-4. Steps shall be numbered sequentially within a task, i.e., task 6-11-10 might have eight steps and they would be numbered 1 through 8. Sequential substeps or other details explaining how to perform the step shall be listed using lower case letters. (Figures 6, 7 and 8).

3.5.7 Tasks. A task is a set of procedural instructions describing a complete start-to-finish, step-by-step maintenance activity. Each separate instruction within a task is a step. Each step shall be numbered sequentially. In multi-page task instructions, supported by a foldout illustration, the associated foldout illustration shall be located on the next right-hand page following the last step.

3.5.8 Text. In addition to the normal requirements of 3.5.7, the following guidance is provided for special situations:

a. General instructions. General instructions for installation of standard hardware may be given.

b. Multi-use instructions. When the instructions for performing an activity can be used for more than one purpose (for example, leading to the removal of a number of components) that activity should be written as a complete function. The function may then be referenced as a required condition in the introduction of other JG functions or included in an appropriate sequence chart.

c. Follow-on maintenance. If, on completion of a function, further maintenance is required to return the system to the required condition, it shall be termed "follow-on maintenance." References to the applicable JG manual, JG function or appropriate sequence chart shall be provided under this heading for instructions to return the equipment to the required condition. (Figure 8).

3.5.9 Illustrations. Illustrations shall convey location and, if applicable, dimensional data with tolerance information. The illustration shall be limited to the equipment applicable to the task performed, plus sufficient surroundings to allow a technician to easily locate the equipment item. When an equipment item is first illustrated in a task, and its location has not yet been specified, a general locator illustration shall be used to identify the location of the equipment item within the system. (Figure 11). Leader lines or sweep arrows shall be used to orient the reader with respect to the equipment and to provide directional orientation in the tasks.

3.5.9.1 Figure numbers. Task number and related illustration number shall correspond, i.e., "Task 2-7-1" and related illustration number "Figure 2-7-1" (Figure 6). Two tasks using the same illustration may be included on the same page with the illustration if the page layout is not crowded. Reference shall be made from the second task to the figure number in the first task. Figure numbers within tasks will not have titles.

3.5.9.2 Task illustration callouts. Callouts shall be provided on the illustration. These callouts will be a number connected by a leader line (with or without arrowheads) to the correct point on the illustration. The callout number shall follow the item description as discussed in the text and be enclosed in parentheses,

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i.e.: "Tighten large turnbuckle (14) by hand until cable is snug." Within each task, illustration callout numbers shall begin with the number 1 and continue consecutively through all illustrations within the task as required regardless of the number of separate illustrations.

3.5.10 Guidelines for typical maintenance functions.

a. Operational checkout. Functions shall include location data for all end items encompassed by the check. All primary equipment and test set turn-on and warm-up procedures shall be included. The expected results shall be followed by the system fault code, if applicable, in parentheses, that directs the technician to the fault isolation procedure used when the expected results are not contained.

b. Removal. All hardware items to be removed shall be specified by name. Instructions shall include matching, marking, or labeling for installation of any component that could be installed incorrectly. This shall be done also for any cable, hose, or line disconnected during removal of any item when the possibility of incorrect installation exists. Access panels, screws/fasteners of different sizes within the same panels shall be identified. Special hardware items (nonstandard fasteners, brackets, or fittings slightly different from others in the same assembly) shall be specified by identifying characteristics (longer, thicker, plastic, etc.). Instructions shall direct when to retain items for reinstallation and to record position of items removed, such as: number of shims at each attaching point.

c. Installation. All hardware items to be installed shall be specified by applicable nomenclature. Instructions shall identify all items retained during removal for reinstallation. When two or more similar items, with same or similar nomenclatures, such as two adapters, two cotter pins, two oils, two lubricants, etc., are to be used or installed within a function, the part number or Government specification shall follow the item nomenclature in parentheses.

Examples:

1. Apply lubricant (MIL-G-28237) to bolt threads.
2. Install cotter pin (MS24665-370) in adjusting bolt.
3. Apply lubricant (MIL-G-46000) to drive shaft.
4. Install locking collar and cotter pin (MS24665-373).

d. Adjustments. Procedures and illustrations indicating method of measurement and/or adjustment shall be provided to return to tolerance such components, assemblies or subsystems as have been determined to be out-of-tolerance (not to include the actual fault isolation).

e. Inspection. Methods, equipment and instructions for normal inspection of components/parts shall be provided. . Allowable service limits and adequate standards for determining when parts should be repaired or replaced shall be stated. Any necessary testing of components or subassemblies shall be

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included. Inspection covered by manuals such as nondestruct inspection (NDI), corrosion control, structural repair, etc., shall not be repeated.

f. Cleaning. Any special cleaning method required shall be described, including the nomenclature and Government specification numbers of the authorized cleaning agents.

g. Lubrication. The type, amount, mode of lubricant application and the Government specification of the lubricant shall be stated. Lubrication points shall be illustrated and adequately identified. The requirements of MIL-HDBK-275 are applicable. Only the lubrication required as a result of the maintenance being performed shall be covered.

h. Servicing. Instructions shall include procedures to check and replenish fuel, oil, hydraulic fluid, other fluids, oxygen, tire pressures and all items involved in completely servicing mobile units, except scheduled equipment lubrication. The information may be in either text, chart or tabular form. Charts or tables shall include tank and reservoir capabilities and identify Government specification number and grade of fuel, oil, fluid and other material used. Specifications and grades shall be shown, grouped on one page or more to facilitate change, as necessary. For fuel, list expansion volume, total fuel capacity, sump capacity, and net fuel capacity for each tank as applicable. For oil, allowance for expansion shall be given. Defueling (depleting) instructons shall be included. The servicing instructions shall be supplemented with diagrams showing location of regular and emergency servicing points.

i. Alignment and adjustment. Step-by-step procedures and illustrations shall be provided for alignment and adjustments following the replacement of parts or assemblies, or as required to determine if the component or equipment meets operational performance standards.

j. Calibration. Any equipment requiring recurring accuracy, checks/adjustment shall be identified and step-by-step procedures given to accomplish these adjustments with recommended equipment for calibration (Portable ATE Calibrator (PATEC), concept is encouraged).

k. Jacking. Jacking procedures shall include a diagram showing the location of points, jack type, and specific methods to be used for jacking. Appropriately illustrated instructions shall be included for hoisting the equipment or major component of the equipment: i.e., missile stages, trailers, antennae, etc. The use of slings and procedures for maintaining correct balance of items to be hoisted will be provided. Leveling instructions shall include location of leveling legs, reference points and access points.

3.5.11 Health hazards. Procedures prescribed for the operating of equipment shall be consistent with the safety standards established by the Occupational Safety and Health Act, (OSHA) Public Law 9-596 and Executive Order 11612. Warnings and cautions shall also be used when hazardous chemicals or adverse health factors in the environment or use of the equipment cannot be eliminated. The contractor shall provide occupational safety, health and fire prevention guidance in the technical data. This guidance will include all applicable safety, health and fire prevention requirements contained in OSHA Standards (Title 29, part 1910 and 1926). If there

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are no applicable OSHA Standards, applicable nationally recognized sources of safety, health and fire prevention standards will be used (i.e., American National Standards Institute) or National Fire codes (published by the National Fire Protection Association). The contractor shall ensure all detailed operations specify the proper safety, health and fire prevention procedures and use of personal protection equipment to adequately protect the worker.

3.6 Job guide index (JX) manual. This manual is applicable to JG manuals only. The JX manual shall contain an alphabetical listing of all systems, subsystems, subsubsystems, equipment nomenclatures, items (LRUs) and activities covered in all the JG manuals cross referenced to the applicable JG manual number. For definition, see 6.3.5. The applicable S/S/SN shall be provided in parentheses following the system, subsystem, subsubsystem, equipment nomenclature or items. For front matter contents, see Table I.

3.7 Fault reporting (FR) manual. The FR manuals shall provide fault reporting information for the operational system faults that are detected by automatic or manual means. The manual shall be prepared for use by experienced maintenance personnel. This manual shall provide coverage of all faults delineated in the Fault Matrix, and shall provide linkage to each system fault tracking manual. For definition, see 6.3.6.

3.7.1 Arrangement. The basic contents and arrangement of the FR manual shall be as follows:

a. Front Matter (Table I)

b. Introduction

c. Section 1. General description

d. Section 2. Fault index and maintenance response data

e. Section 3. Maintenance planning

f. Section 4. Flow diagrams

3.7.2 Sections. Sections will be arranged as follows:

3.7.2.1 Section 1. Section 1 shall contain a general description of the maintenance concept, planning, response, flow diagrams, system function and information regarding methods and procedures of fault identification, fault coding and fault matrix (root mode) concept. This section shall also provide a listing of fault codes and their meanings.

3.7.2.2 Section 2. Section 2 shall contain the fault index and maintenance response data. This section will provide a fault index, fault response procedures and contain a reference to the common maintenance diagrams in Section 4. The fault index shall provide a cross reference between fault indication(s), fault numbers and the fault response figure number. The fault index shall be keyed to specific groups of fault indications, such as, local mode faults, printouts with or without lamp status, and miscellaneous faults. The maintenance response figure(s) shall be divided into

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four areas of data: (1) fault indications; fault number, status, evaluation, and when discovered; (2) dispatch data, technical orders, operational ground equipment/maintenance ground equipment (OGE/MGE) and maintenance actions necessary to correct the malfunction(s); (3) system impact(s); and, (4) a brief and concise fault description. These data categories are the core of the fault report reporting manual and they must provide continuity to other documents used to maintain the weapon system.

3.7.2.3 Section 3. Section 3 shall contain maintenance planning requirements and include data required to effect the most efficient use of teams and equipments. This section will provide a list of support equipment (SE), required to repair the weapon system, the team designations, and a time chart for performing various maintenance tasks which also correlates the optimum team to perform a specific maintenance function.

3.7.2.4 Section 4. Section 4 shall contain maintenance flow diagrams and replaceable items for common functions that are not contained elsewhere in this manual. This section will also provide simple block diagrams that sequences an experienced maintenance operator through a fault repair procedure. These flow diagrams will be numbered and referenced in Section 2.

3.7.2.5 Classified material. No classified material shall be included in this manual. Any such material shall be contained in a classified supplement.

3.7.2.6 Text. The FR manual shall provide operational system fault identification-diagnostic and reporting instructions. It shall provide the appropriate operations or maintenance personnel with initial fault indication descriptions, instructions to locate any appropriate fault diagnostic or isolation procedures, and shall list fault codes which identify specific maintenance responses. Tables consisting of fault matrix data shall include information appropriate to the specific weapon system equipment.

3.7.2.7 Page size. The FR manual page size shall be 8-1/2 by 11 inches.

3.8 Fault tracking (FT) manual. The FT manuals provide system troubleshooting data to maintenance personnel. The FT manual shall include all troubleshooting for the system being addressed and shall have a direct interface with the FR manual for those faults addressed in the FR manual. For definition, see 6.3.7. Faults identified by the automatic test shall be covered to the extent defined by 3.7.

3.8.1 Tests. The FT manual shall provide troubleshooting data to rectify faults identified and coded by the fault reporting manual and shall provide fault description and FT procedures for specific faults identified by any other method.

a. Automatic or self-tests. If automatic or self-test capabilities of existing maintenance support equipment are different or more extensive than those in the automatic fault reporting system, they will also be used to extend the FT processes.

b. Indications. Automatic checkout indications (lamps, printout, displays, etc.) shall be used as either interim or end results to develop the FT process.

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3.8.1.1 Arrangement of subject matter. All fault identification, description and isolation information shall be arranged in alphanumeric sequence within each system/subsystem/subject. An FT procedure shall be provided for each specified fault or reference shall be made to an applicable isolation procedure contained elsewhere in the manual.

3.8.2 Manual content. The FT manual shall contain sufficient information to enable a technician to troubleshoot systems and subsystems to a level consistent with the approved maintenance concept. Procedure-type references shall be limited to functions within the JG manual series. The manual shall be prepared to enable a qualified technician, considered to possess pertinent knowledge and practical abilities, to perform the designated tasks. The manual shall identify, describe and isolate all faults identified and reported by the fault reporting manual, and shall provide FT procedures for specific faults identified by other methods. With the exception of the FR manual, other FT manuals, JG manuals, and the wiring data and schematic diagrams manuals, the FT manual shall require minimal support from other manuals. The FT manual shall provide troubleshooting for all faults that are observable during all phases of system operation and maintenance activities except as follows: a) those faults isolated by automatic test equipment during normal operations when test equipment isolates the fault to a single LRU; b) those diagnostic processes covered by the FR manual.

3.8.3 Page size. The FT manual page size shall be 8-1/2 by 11 inches.

3.8.4 Arrangement. Unless otherwise specified by the procuring activity, separate FT manuals shall be prepared for each system, subsystem or subsubsystem as appropriate. The FT manual shall be arranged in the following manner:

a. Front Matter (Table I)

b. Introduction

c. Section 1 of each FT manual shall contain general information pertinent to fault isolation.

d. Section 2 shall contain a description of the fault code system and a fault code index.

e. Subsequent sections shall be arranged in the following sequence:

1. Fault identification and description.

2. Access and locator data.

3. Supplemental data, as required.

4. Fault isolation procedures - operation checks, operating parameters.

5. Schematic diagrams.

3.8.4.1 Fault identification. All fault inputs from the FR manual shall be identified by the code identified in the FR manual. Faults from other sources shall be identified by S/S/SN and a sequence number.

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3.8.4.2 Fault description. Fault description shall consist of a series of words and/or phrases arranged in any of the formats described in 3.8.4.4.3. This format lists fault indications together with related observations, conditions, operations performed and other pertinent information. Fault description information for faults defined by the FR manual shall restate the information contained therein.

3.8.4.3 Access and location information. The location of isolation points unique to the isolation process, except where location is obvious or becomes obvious as a result of gaining access, shall be shown or described. The means and location for gaining access to components shall be provided except where obvious. This information shall be provided in either illustrated or written form. Written access or location information shall be limited to avoid extensive description. Several illustrations may be included on a page provided readability is maintained. (Figure 12).

3.8.4.4 Fault isolation (troubleshooting) procedures. An FT procedure shall be provided for each identified fault in the fault index. The isolation procedure may be used for more than one identified fault provided the faults are concisely and adequately isolated by one procedure. Isolation procedures shall consist of appropriate fault codes, and shall direct the technician through the appropriate actions and to the proper instructions for correcting the fault. Fault isolation shall be performed to the level established by the approved maintenance concept.

3.8.4.4.1 Reference. Each fault tree or fault isolation procedure that does not complete the fault isolation process within the system being covered, shall reference the interfacing FT manual, by fault code, for continuation of the fault isolation process. For example: if a fault was reported concerning loss of ac power at the Cluster Maintenance Facility (CMF) Security Alarm Set (50-CA-00), the fault tree would start by checking fuses, power switches and power interface connectors contained in 50-CA-00 and then reference to the CMF power set (40-KA-00). Each statement shall be explicit and positive to insure the mechanic understands what to do, and what results to expect. Subsequent blocks shall consider all outputs to be expected to result from preceding blocks. Reference shall be made to other maintenance data S/S/SN number or TO number in areas where the technician may require additional guidance in order to accomplish directed tasks.

3.8.4.4.2 Methods. The recommended maintenance actions shall be specific for the reported malfunction and normally will not call for any fault diagnosis to be repeated that has been carried out by the automatic fault isolation system. However, the manual shall direct the accomplishment and shall define equivalent diagnostic procedures in the event fault diagnosis has not been carried out. Fault isolation procedures shall be provided to permit the use of authorized test equipment, if available, to reduce or improve the fault isolation process as established by the maintenance concept.

3.8.4.4.3 Format. Fault isolation format selection guide and troubleshooting method guide are shown in figures 13 and 14. Typical fault isolation (troubleshooting) formats are shown in figure 15 through 19. These may be used as guidelines in developing troubleshooting procedures. Depending on the type of equipment or system being analyzed, select a format most suited to that system or equipment.

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3.8.4.5 Fault tracking manual schematic diagrams. Reference will normally be made to schematics in the schematic diagram manual. However, upon specific approval of the procuring activity, schematics may be duplicated in the FT manual if they will improve the use of that manual.

3.9 Wiring data (WD) manual. The WD manuals provide wiring data for the M-X weapon system. For definition, see 6.3.8. This manual shall be prepared in multi-manual (sectionalized) format and utilize the contractor's engineering data if in compliance with the requirements of MIL-M-38784. Refer to 3.11.

3.9.1 WD manual format. The WD manual shall be prepared in sectionalized format to coincide with the structure of the JG sets. Each manual shall be arranged as follows:

a. Front Matter (Table I)

b. Section 1 Introduction

c. Section 2 Equipment List

d. Section 3 Wire and Connection List

e. Section 4 and subsequent sections, Wiring Diagrams

3.9.1.1 Section 1, WD manual introduction. This section shall contain the introduction portion of that particular WD manual. In addition to the requirements of MIL-M-38784, the introduction shall contain information applicable to the following specific requirements:

a. Configuration(s) covered. A listing of the models, types and series of items if more than one configuration is covered.

b. Data accessing system. Test and illustrated explanation shall describe:

How to find the applicable wiring diagrams/data.

How to find the wiring effectivity.

How to use the wire list to determine wire replacement information such as wire type, length, method of termination, crimping tool used, termination point, etc.

c. Symbols and abbreviations. A explanation of system unique symbols and abbreviations used in the WD manual shall be provided. Illustrations of the symbols used in the WD manual shall be provided. Reference to the appropriate MIL-STD/ANSI/USAS/IEEE standards shall be made.

d. System, subsystem and subsubsystem explanation. An explanation of the S/S/SN system with applicable titles shall be provided.

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3.9.1.2 Section 2, WD manual equipment list. This section shall contain the part numbers of each unit described in the WD and schematic diagram (SD) manuals with the applicable higher level reference designations.

3.9.1.3 Section 3, WD manual wire and connection list. These sections shall contain a wire harness list and a connection list.

3.9.1.4 Section 4 and subsequent sections, WD manual wiring diagrams. These sections shall contain point-to-point interconnection wiring diagrams and associated charts.

3.9.1.4.1 Table of contents. Each section shall include table of contents pages which shall list the data/diagrams within the section by subject nomenclature and number.

3.9.1.4.2 Index of effective diagrams. The index of effective diagrams shall list diagram applicability to equipment configuration.

3.9.1.4.3 Wiring diagrams/wire lists. Point-to-point interconnection wiring diagrams or associated charts shall be provided.

3.10 Schematic diagram (SD) manual. The SD manual shall provide schematics for the M-X weapon system. For definition, see 6.3.9. This manual shall be prepared in multi-manual (sectionalized) format and utilize the contractor's engineering data if in compliance with the requirements of MIL-M-38784. Refer to 3.12.

3.10.1 Schematic diagrams. The purpose of schematics is to provide information to the technician to understand the troubleshooting process followed in the manual, and to assist in establishing troubleshooting procedures for faults not eliminated through use of the fault isolation procedures. Schematics shall be provided for those systems whose functions are not of general knowledge, are extremely complex, or cannot be determined by visual inspection. Electrical, hydraulic, pneumatic and associated mechanical elements shall be presented in schematic format.

3.10.2 SD manual format. The SD manual shall be prepared in sectionalized format to coincide with the structure of the JG sets. Each manual shall be arranged as follows:

a. Front Matter (Table I)

b. Section 1, Introduction

c. Section 2 and subsequent sections, System Schematic Diagrams

3.10.2.1 Section 1, SD manual introduction. The first section shall contain the introduction to that particular SD manual including the introduction portion applicable to system schematic diagrams. In addition to the requirements of MIL-M-38784 the introduction shall contain information applicable to the following specific requirements:

a. Configuration covered. A listing of the models, types and series of items being covered if more than one configuration is covered.

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b. Data accessing system. Text and illustrated explanations shall describe:

How to find the applicable system schematic diagram contained in Section 2 and subsequent sections.

How to find the system schematic diagram effectivity.

The purpose and use of system schematic diagrams.

c. Symbols and abbreviations. An explanation of the system unique symbols and abbreviations used in the SD manual shall be provided. Illustrations of the symbols used shall be included. Reference to the appropriate MIL-STD/ANSI/USAS/IEEE standards shall be made.

d. System, subsystem and subsubsystem explanation. An explanation of the S/S/SN system with applicable titles shall be provided.

3.10.2.2 Section 2 and subsequent sections. The sections shall contain the system schematic diagrams.

a. Tables of contents. Each section shall include table of contents pages which shall list the data/diagrams within that section by subject nomenclature and number.

b. Index of effective diagrams. The index of effective diagrams shall list diagram applicability to equipment configuration.

3.11 Wiring data index (WX) manual. The WX manuals list all wiring data drawings or lists contained in the system WD manual. For definition, see 6.3.10. Refer to 3.9 for contents. For Front Matter, see Table I.

3.12 Schematic diagram index (SX) manual. The SX manuals list all schematic diagrams contained in the system SD manual. For definition, see 6.3.11. Refer to 3.10 for contents. For Front Matter, see Table I.

4. QUALITY ASSURANCE PROVISIONS

4.1 Requirements. The quality assurance provisions shall be in accordance with MIL-M-38784 and MIL-P-38790.

5. PACKAGING

5.1 Packaging requirements. Packaging, packing and marking for shipment shall be in accordance with MIL-P-38790.

6. NOTES

6.1 Intended use. Technical manuals prepared in accordance with this specification are to be used for organizational and intermediate level maintenance of installed systems and equipment for M-X.

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6.1.1 Organizational maintenance. The organizational maintenance concept is to remove and replace at the LRU level, either on a scheduled basis as determined from design characteristics, or on an unscheduled basis as determined from instrumentation, sampling, trend analysis, condition monitoring, inspection, or malfunction. Repair in place will be accomplished only when justified by the results of maintenance analysis. The equipment maintenance concept is to remove and replace LRUs if possible. LRUs removed at the organizational level will be repaired at the level and location dictated by the optimum repair level analysis (ORLA).

6.1.2 Intermediate maintenance. Intermediate maintenance will be performed at the operating base (OB), designated assembly area (DAA), in shops, laboratories, and special areas assigned to accommodate maintenance actions such as component repair, major inspection, calibration, and buildup/replacement of major aerospace vehicle equipment (AVE) components. The identification of specific intermediate repair capabilities at the OB and DAA will be determined by ORLA.

6.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Para No.	Data Requirement Title	DID No.	Option
3.2 thru 3.12	Technical Orders	DI-M-3407	None

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 General weapon (GW) manual. This manual describes the overall system and provides a brief description of systems and subsystems. It also provides an overview of the organizational and intermediate maintenance manual sets, their structures, arrangements and how the sets are to be used. See 3.2 for detail requirements.

6.3.2 General system (GS) manual. These manuals provide detailed system, subsystem, and subsubsystem description, theory of operation, type, main features, illustrations, and support data necessary to complement the job guide manuals. See 3.3 for detail requirements.

6.3.3 Sequence chart (SC) manual. This manual provides sequence requirements charts for complex or sequence critical maintenance activities. See 3.4 for detail requirements.

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6.3.4 Job guide (JG) manual. These manuals provide detail procedures for organizational and intermediate level maintenance. They are a step-by-step, highly illustrated, "how to" sets of instructions. See 3.5 for detail requirements.

6.3.5 Job guide index (JX) manual. This manual contains an alphabetical listing of those items covered in the JG manuals. See 3.6 for detail requirements.

6.3.6 Fault reporting (FR) manual. This manual provides a means of conveying fault reporting information between the operating crew and maintenance personnel. The manual contains operational fault identification and reporting instructions as determined by the automatic fault detection/isolation system. See 3.7 for detail requirements.

6.3.7 Fault tracking (FT) manual. These manuals provide troubleshooting data to rectify faults identified and coded by the fault reporting manual, and provide fault descriptions and fault isolation procedures for specific faults identified by methods other than the automatic system. See 3.8 for detail requirements.

6.3.8 Wiring data (WD) manual. These manuals provide complete wiring data for the system excluding the schematics contained in the schematic diagram manual. See 3.9 for detail requirements.

6.3.9 Schematic diagram (SD) manual. These manuals provide complete system/subsystem schematics prepared and presented on three level format (block, simplified and detailed schematic). All schematic diagram manuals are based on data as released by engineering. See 3.10 for detail requirements.

6.3.10 Wiring data index (WX) manual. This manual is a composite index of all wiring data drawings or lists contained in all wiring data manuals. See 3.11 for detail requirements.

6.3.11 Schematic diagram index (SX) manual. This manual is a composite index of all schematic diagrams contained in all schematic diagram manuals. See 3.12 for detail requirements.

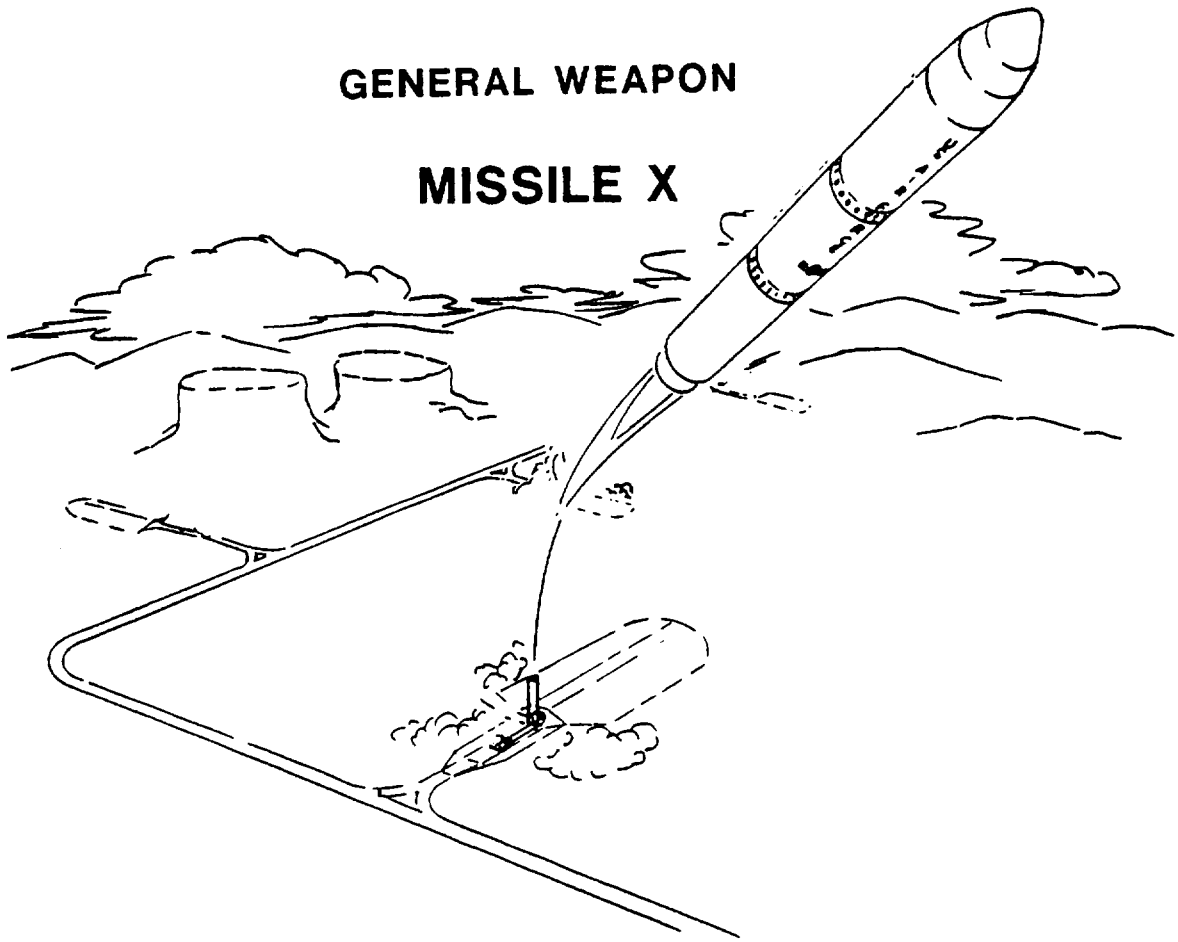
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TO 21-MG118A-M00GW-AA1

TECHNICAL MANUAL

GENERAL WEAPON

MISSILE X



(Manual prepared by ABC Aerospace Co.)
FXXXXX-X-XX-XXXX

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FIGURE 1. Sample title page, general weapon manual.

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WARNING/SAFETY SUMMARY

WARNING

(Specific warning information related to the overall system or equipment covered within the manual shall be noted here.)

(Separate paragraphs, as required, to address those separate hazard activities or areas not covered by specific warnings within the text of the manual.)

Definitions: WARNINGS, CAUTIONS and NOTES used in this manual have the following signifiance:

WARNING

An operating or maintenance procedure, practice, condition, statement, etc., which, if not strictly observes, could result in injury to or death of personnel.

CAUTION

An operating or maintenance procedure, practice, condition, statement, etc., which, if not strictly observes, could result in damage to, or destruction of, equipment or loss of mission effectiveness.

Note

An essential operating of maintenance procedure, condition, or statement which must be highlighted.

SAFETY PRECAUTIONS

HAZARDOUS EQUIPMENT OR MATERIAL	LOCATION	SAFETY ACTION REQUIRED	REASON

FIGURE 2. Sample warning/safety summary page format.

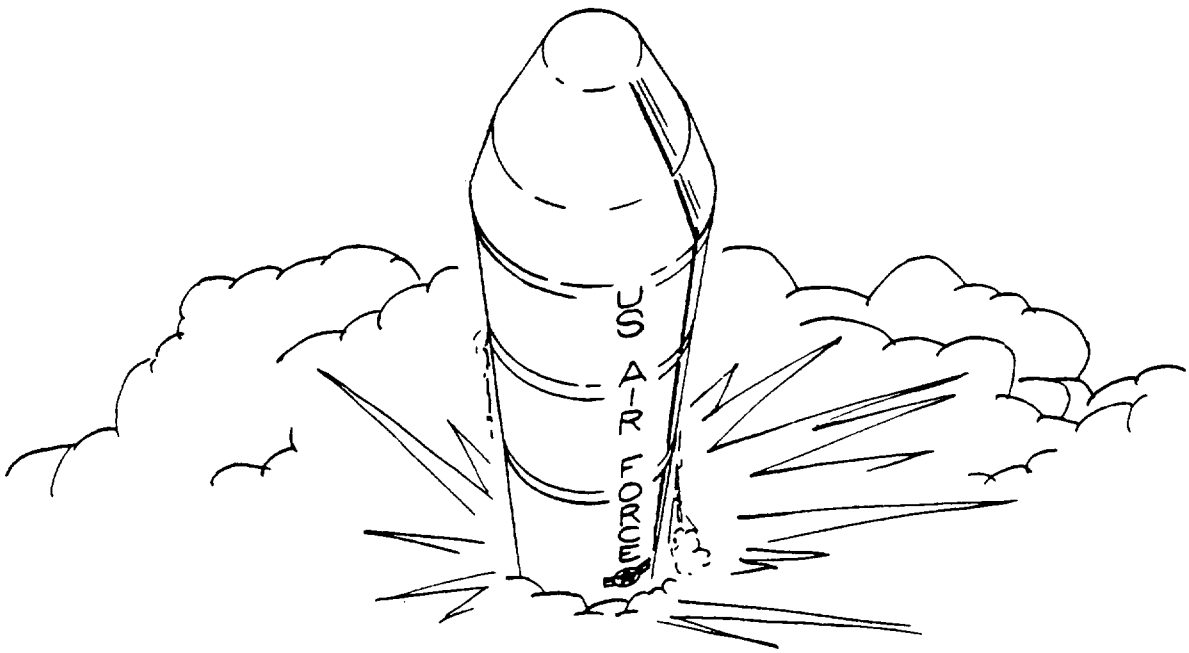
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TECHNICAL MANUAL

GENERAL SYSTEM

MISSILE X



(Manual prepared by ABC Aerospace Co.)
FXXXXX-X-XX-XXXX
FYYYYY-Y-YY-YYYY

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FIGURE 3. Sample title page, general system manual.

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TECHNICAL MANUAL

JOB GUIDE INDEX

(Manual prepared by ABC Aerospace Co.)
FXXXXX-X-XX-XXXX

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FIGURE 4. Sample title page, job guide index.

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TO 21-MG118A-M80JG-AA1

TECHNICAL MANUAL

JOB GUIDE

TRANSPORTATION AND HANDLING MISSILE STAGES I, II, AND III

(Manual prepared by ABC Aerospace Co.)
FXXXXX-X-XX-XXXX

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Published under the authority of the Secretary of the Air Force.

1 MAY 1981

FIGURE 5. Sample title page, job guide series.

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2-7 REMOVE STAGE I FROM CONTAINER (80-AA-07)

INPUT CONDITIONS

APPLICABILITY: All

REQUIRED CONDITIONS: Refer to TO 21-MG118A-MOOSC-RAI

SUPPORT EQUIPMENT and SPECIAL TOOLS:

Part No. (Mfg. Code)	Nomenclature	Alternate	Application
6000218 (04236)	Pneumatic Control Cart	None	Task 2-7-3 and 2-7-5
9001234 (04236)	Grounding Cable and Reel Set	None	Task 2-7-2 through 2-7-7
2727272 (04111)	Cable Winch	None	Task 2-7-3 and 2-7-6
Coml.	Step Ladder	N/A	Task 2-7-3

SUPPLIES: (Consumables):

Nomenclature	Specification or Govt Std No.	Part No. (Mfg. Code)	Application	Qty.
Wipe Rags	–	Coml	Task 2-7-1 and 2-7-6	AR

WARNING

Stages must be grounded at all times or ignition can occur. If a stage ignites, serious injury or death will result.

2-7-1 PREPARE CONTAINER AND RECEIVING AREA FOR STAGE I REMOVAL

1. If needed, turn on apron floodlighting.
2. Verify receiving apron (4) and facility rails are free of obstructions.
3. Check stage container aft end for loose, missing, or damaged parts:
   a. Doors (2 and 3).
   b. Hinges (1).
   c. Handles (5).
   d. Seal (8) on door jamb.
   e. Security lock (6) and seal (7).

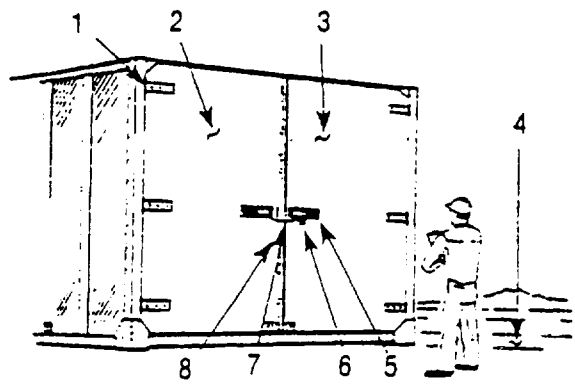


Figure 2-7-1

FIGURE 6. Job guide function input conditions page sample.

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TO 21MG118A-M80JG-AA1

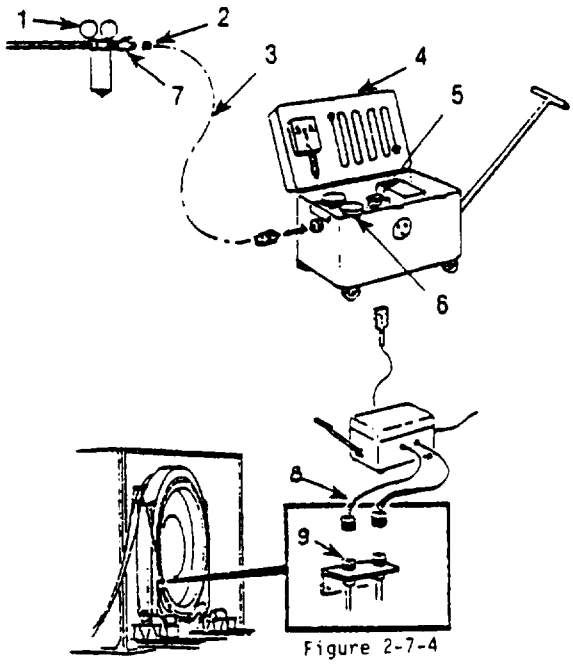


Figure 2-7-4

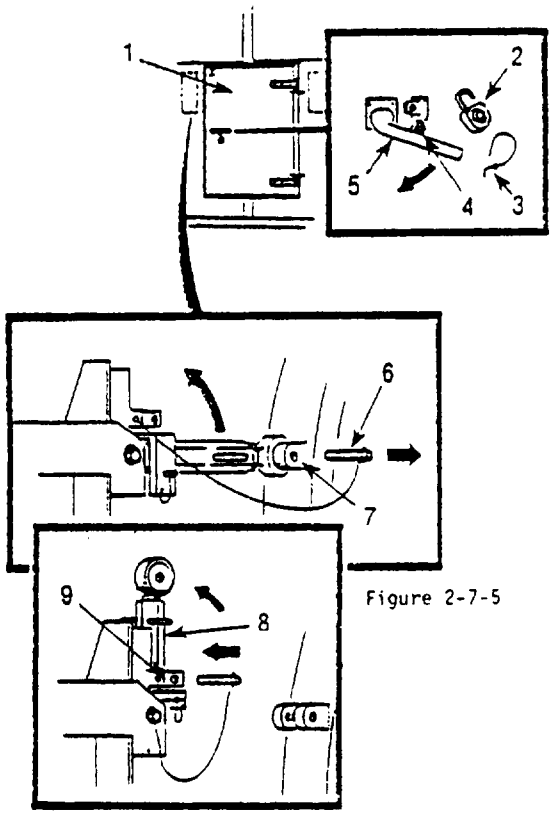


Figure 2-7-5

2-7-4 CONNECT PNEUMATIC CONTROL CART BETWEEN FACILITY AND STAGE I END RING

1. Connect Pneumatic Control Cart (4) to facility air supply (1):
   a. Verify facility air-pressure gage indicates between 80 and 100 psig.
   b. Push hose connector (2) on cart hase (3) onto air nipple at facility air station.
2. Remove moisture and foreign material from cart hose:
   a. Open bleed valve (5) on cart (4).
   b. Open regulator valve (7) on facility air supply.
   c. Close bleed valve (5) on Cart.
3. Verify air gage (6) on cart indicates between 80 and 100 psig.
4. Connect cart air supply hose (8) to end ring air connectors (9).

2-7-5 RELEASE STAGE I RESTRAINT

1. Open access door (1) on stage container forward end:
   a. Open security lock (2), remove from hasp and snap lock closed on hasp latch (4).
   b. Remove inspection seal (3) from door.
   c. Turn door handle (5) in direction of arrow and open door fully.
2. Remove two red retaining pins (6) from strut clevises (7). If pins do not release, loosen turn buckle (8).
3. Turn strut (7) up to stowed position until strut clevises are seated into stowing trunions (9).
4. Push retaining pin (6) into stowed trunion (9) until pins are seated fully.
5. Close and lock access door (1).

80-AA-07
2-54

FIGURE 7. Typical job guide text/art page.

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2-7-6 TRANSFER STAGE I FROM CONTAINER TO FACILITY

1. Open facility door to full height.
2. Turn direction switch (1) on Pneumatic Control Cart pendant (2) to AFT Position.

WARNING

Make sure personnel are clear of rails. Failure to keep personnel clear of rails and moving stage may result in serious injury.

Make sure C-clamp and grounding cable of reel set are grounded at all times. Failure to maintain good electrical ground between stage and facility may cause ignition.

CAUTION

If aft end passes beyond yellow alignment marks on floor near rails, end ring may be damaged.

Note

Maximum speed of stage is about an inch per second. Stage may be slowed down by releasing trigger slightly.

3. Press and hold trigger (3) on cart pendant (2) while verifying stage moves out of container and into facility.

4. Stop stage motion by releasing trigger (3) on pendant (2) when mark (4) on aft truck frame (5) lines up with mark (6) on floor.

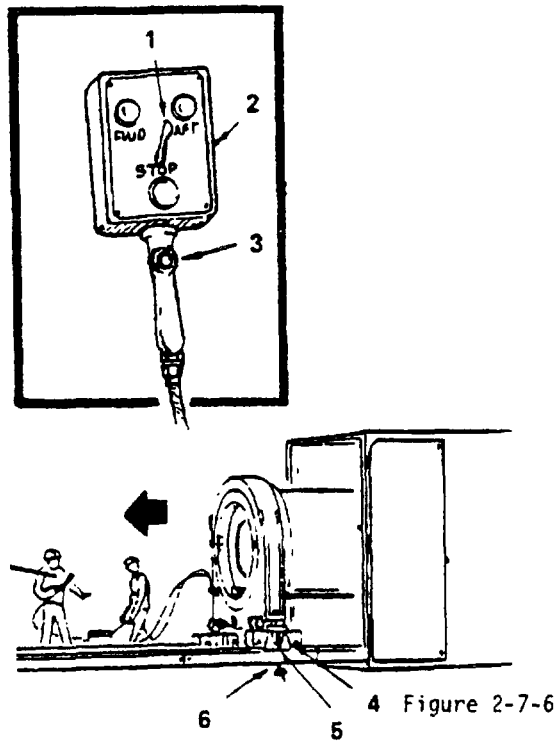


Figure 2-7-6

FOLLOW ON MAINTENANCE:
Refer to : TO 21-MG118A-MOOSC-RA1

******

FIGURE 8. Typical job guide text and art with follow-on instructions.

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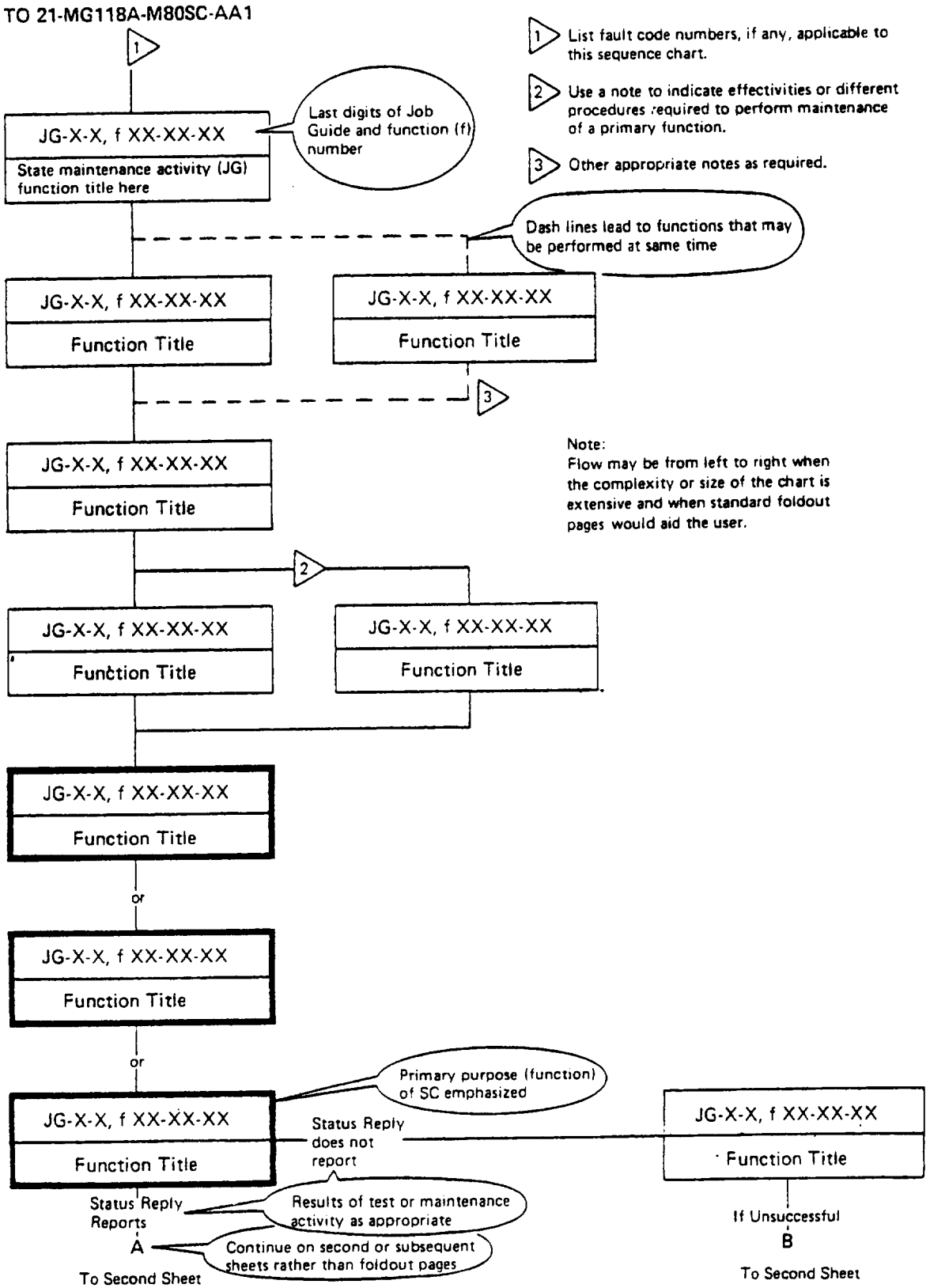


FIGURE 9. Sample sequence chart format

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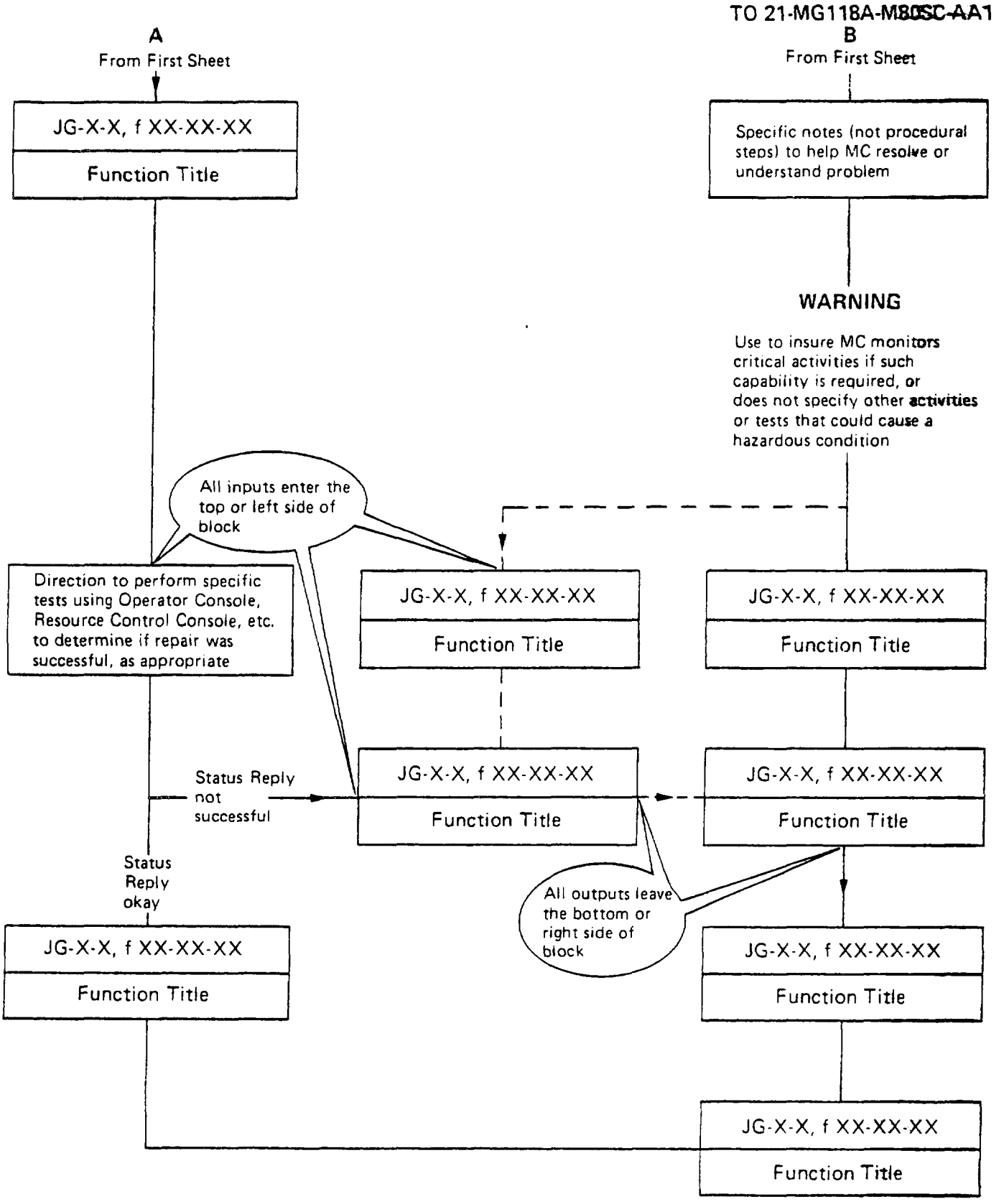
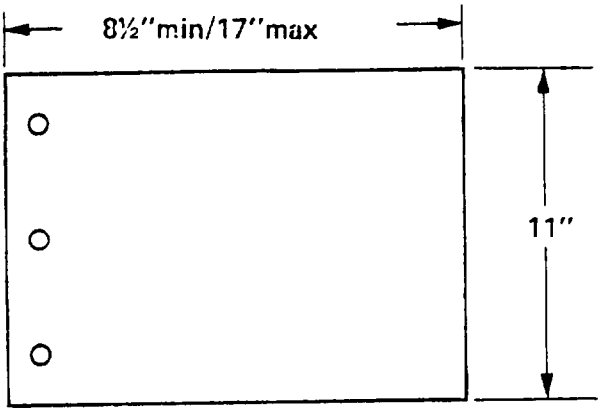
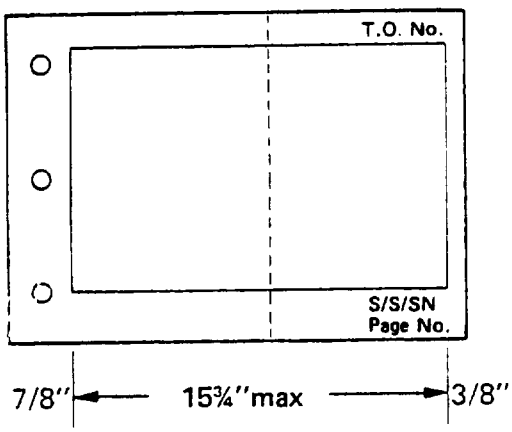


FIGURE 9. Sample sequence chart format - Continued

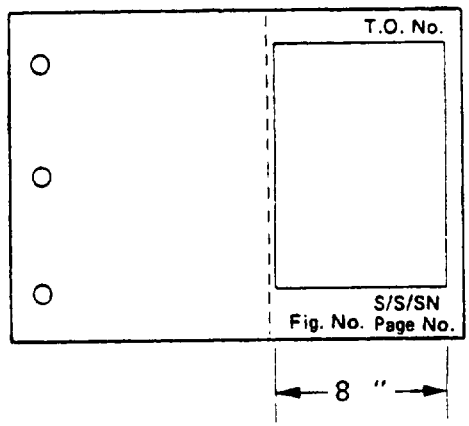
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Page Size



Full Illustration



Blank Apron

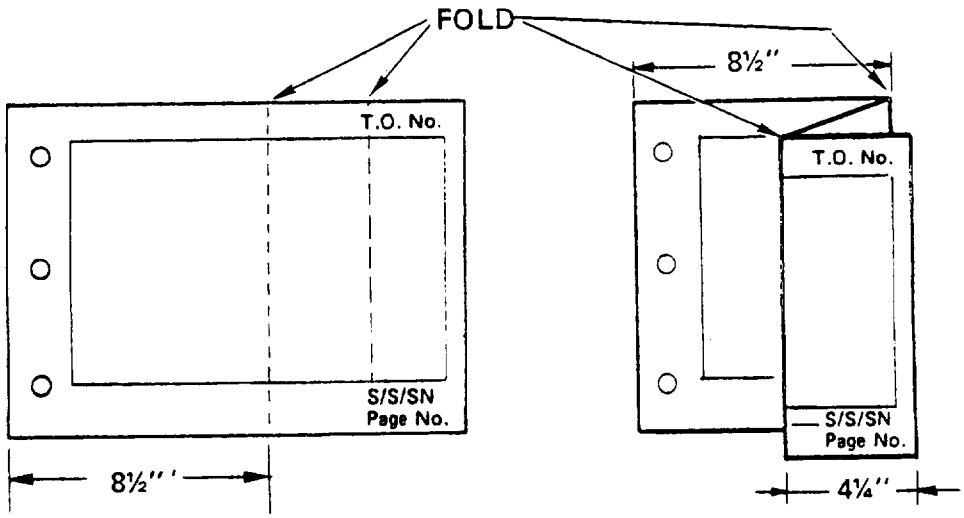
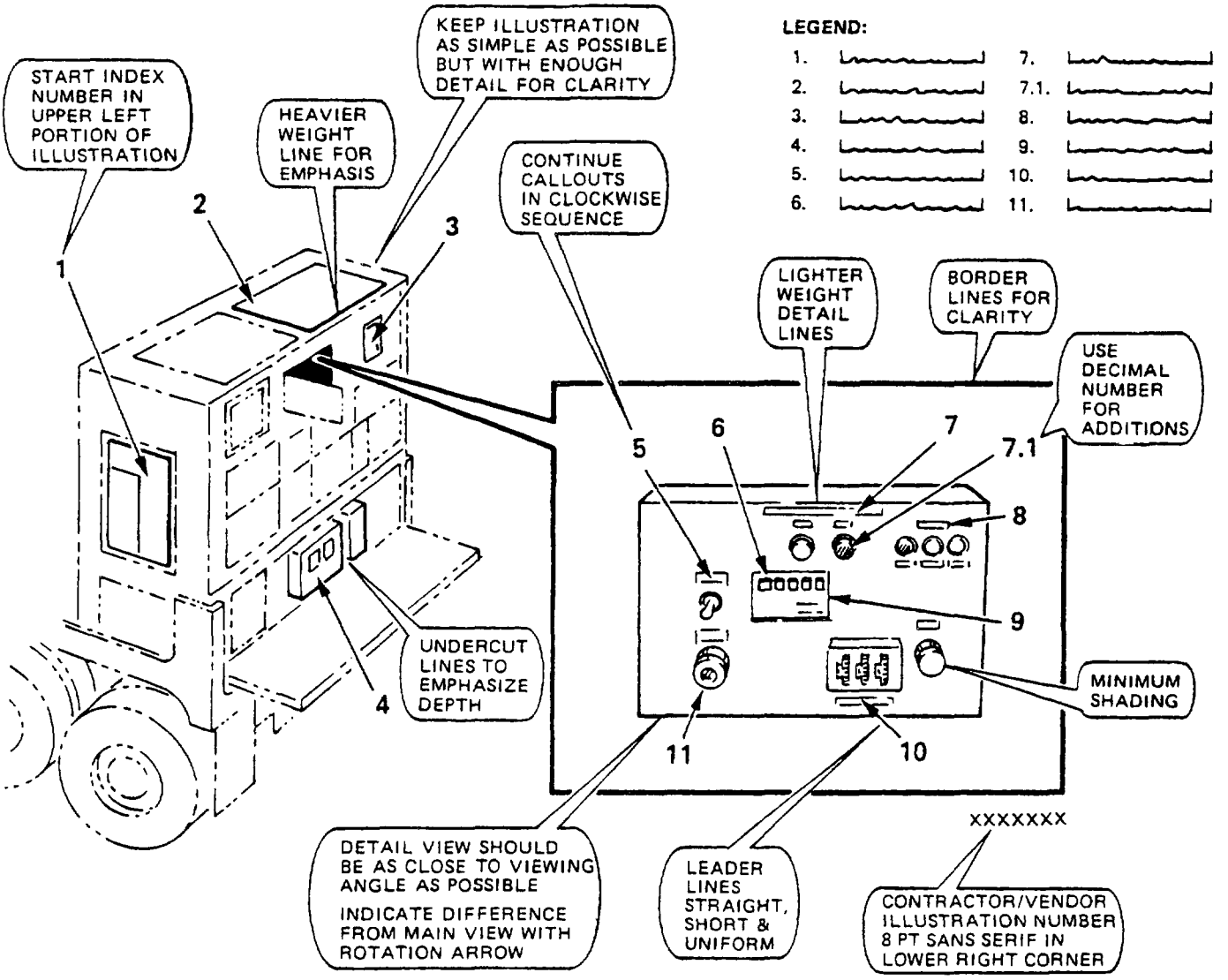


FIGURE 10. Job guide foldout page format.

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NOTES:

- ALL ILLUSTRATIONS WILL BE LINE DRAWINGS.
- ILLUSTRATIONS SHOULD APPEAR ON SAME PAGE OR AS CLOSE AS POSSIBLE TO APPLICABLE TEXT.
- KEEP ILLUSTRATION SIMPLE, BUT WITH ENOUGH DETAIL TO AMPLIFY TEXT. UNDERCUTTING OF LINES SHOULD BE USED TO INDICATE DEPTH AND ADD DEFINITION TO DRAWINGS.
- DO NOT REPEAT ILLUSTRATION WITHIN A JOB GUIDE FUNCTION UNLESS CLARITY OR USE WILL BE SUBSTANTIALLY AIDED. FOR EXAMPLE – TO AVOID AN ILLUSTRATION CLUTTERED WITH TOO MANY INDEX NUMBERS.
- DETAILED ITEMS SHOULD BE SHOWN FROM A NORMAL VIEWING POSITION. IF DETAIL VIEW IS USED THAT IS NOT SAME VIEWING ANGLE AS LOCATOR DRAWING, USE ROTATION ARROW. FOR CLARITY, USE BORDER AROUND DETAIL VIEW.
- SHADING ON PICTORIAL ILLUSTRATIONS SHOULD BE MINIMIZED.
- CALLOUT NUMBERS SHALL BE TYPESET OR PREPARED BY MECHANICAL METHODS, RATHER THAN FREEHAND LETTERING. TYPE SHALL BE BOLD FACE SANS SERIF.
- AVOID DIRECT NOMENCLATURE CALLOUTS WHEN POSSIBLE BY USING CALLOUT NUMBERS AND LEGENDS IF REQUIRED. USE MEDIUM WEIGHT SANS SERIF CAPS.
- CALLOUT NUMBERS SHOULD START IN UPPER LEFTHAND PORTION OF ILLUSTRATION AND, IN SEQUENCE, PROCEED CLOCKWISE AROUND MAIN ILLUSTRATION, BEFORE CONTINUING AROUND DETAIL VIEW. WHEN CHANGING ART, TO ADD A NUMBER BETWEEN EXISTING NUMBERS, USE A DECIMAL NUMBER (EXAMPLE – 1.1 BETWEEN 1 AND 2).
- CALLOUT NUMBERING SHALL BE SEQUENTIAL FROM ONE ILLUSTRATION TO THE NEXT WHEN CONTAINED WITHIN ONE JOB GUIDE TASK.
- ILLUSTRATIONS MAY BE INCLUDED ON TABLES AND CHARTS WHERE NECESSARY FOR READER UNDERSTANDING.
- LEADER LINES SHOULD BE STRAIGHT, SHORT AND UNIFORM. THEY SHOULD NOT DOG-LEG UNLESS NECESSARY, AND SHOULD NOT CROSS EACH OTHER OR OBSCURE ESSENTIAL DETAIL.
- ALL LEADER LINES MAY BE HIGHLIGHTED ON ONE OR BOTH SIDES, WITH ARROWHEADS PREFERRED.
- PHANTOM LINES MAY BE USED TO REFERENCE STRUCTURE OR ITEMS NOT LISTED OR INDEXED.

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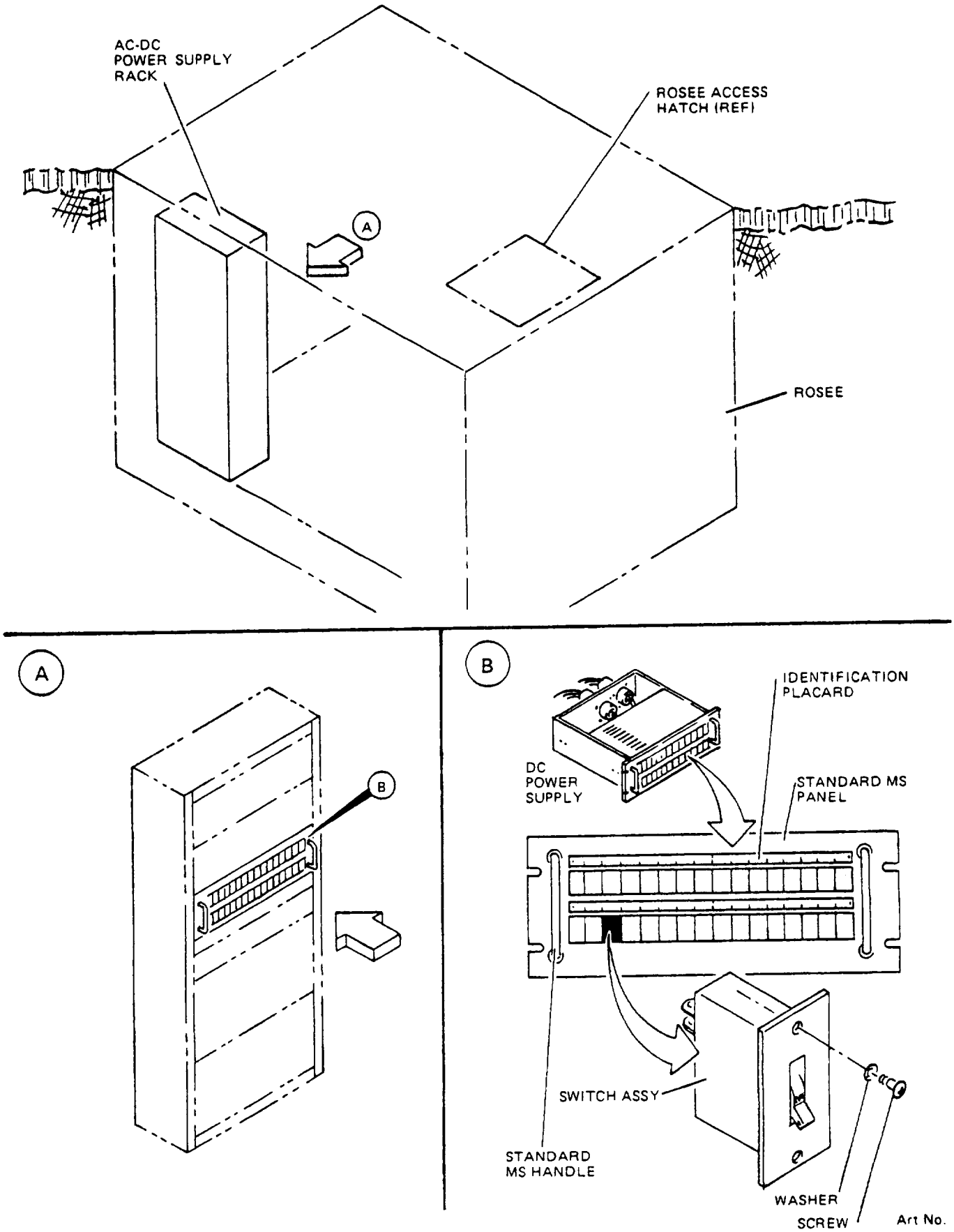


FIGURE 12. Fault isolation access and location information sample.

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Fault Isolation Format	Advantages	Limitations
Tabular (Fig. 17, 18, 19)	• Compact on page • Can be produced on preprinted page • Easy to use	• Procedural flow might not be as easy to follow
Branching (Fig. 16)	• Procedural flow easy to follow • Can use computer to generate	• Requires more space than tabular
Narrative (Fig. 15)	• Easy to write and produce	• Requires more time to use - might be tedious • Procedural flow not as easy to follow

FIGURE 13. Fault isolation format selection guide.

TROUBLESHOOTING METHOD GUIDE / SELECTION CRITERIA	HARDWARE COMPLEXITY		QUANTITY PRODUCED		BUILT-IN TEST		NO. OF PERSONNEL		PERSONNEL EXPERIENCE/TRAINING		TASKS		EXTENT OF REPLACEMENT			MISSION IMPORTANCE		FREQUENCY OF OCCURRENCE		ENVIRONMENT		WORKING AREA	
	Simple	Complex	Few	Many	Little	Extensive	Few	Many	Little	Extensive	Simple	Complex	Unit, Assembly, Module	Component Part		Unimportant	Essential	Rare	Common	Sheltered	Exposed	Ample	Limited
LOGIC TREES (Tabular or Narrative)		X		X	X	X		X	X			X	X				X		X	X	X	X	X
TROUBLESHOOTING DIAGRAMS	X	X	X	X	X		X	X		X	X		X	X		X		X	X	X		X	

FIGURE 14. Troubleshooting method guide.

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FAULT SYMPTOM INDEX

ITEM / MALFUNCTION OR SYMPTOM	FAULT CODE
ENGINE	
Fails to start but turns over	XX-XX-XX
Does not reach operating rpm	XX-XX-XX
[illegible]	XX-XX-XX
[illegible]	XX-XX-XX
GENERATOR	
[illegible]	XX-XX-XX
[illegible]	XX-XX-XX
[illegible]	
[illegible]	
STEERING	
[illegible]	
[illegible]	
[illegible]	
[illegible]	

FIGURE 15. Fault symptom index format guide.

SYMPTOM INDEX NO. TEST OR INSPECTION CORRECTIVE ACTION	FAULT CODE
ENGINE	
1. ENGINE FAILS TO START	
Step 1. Check fuel tank level and assure crossover valve lever to MAIN TANI	
If step 1 checks okay, go to step 2.	
Step 2. Remove [illegible]	
Disconnect [illegible]	
Spray [illegible]	
a. If engine fails to start [illegible]	
b. If engine starts [illegible]	XX-XX-XX
[illegible]	

FIGURE 16. Narrative logic tree format guide.

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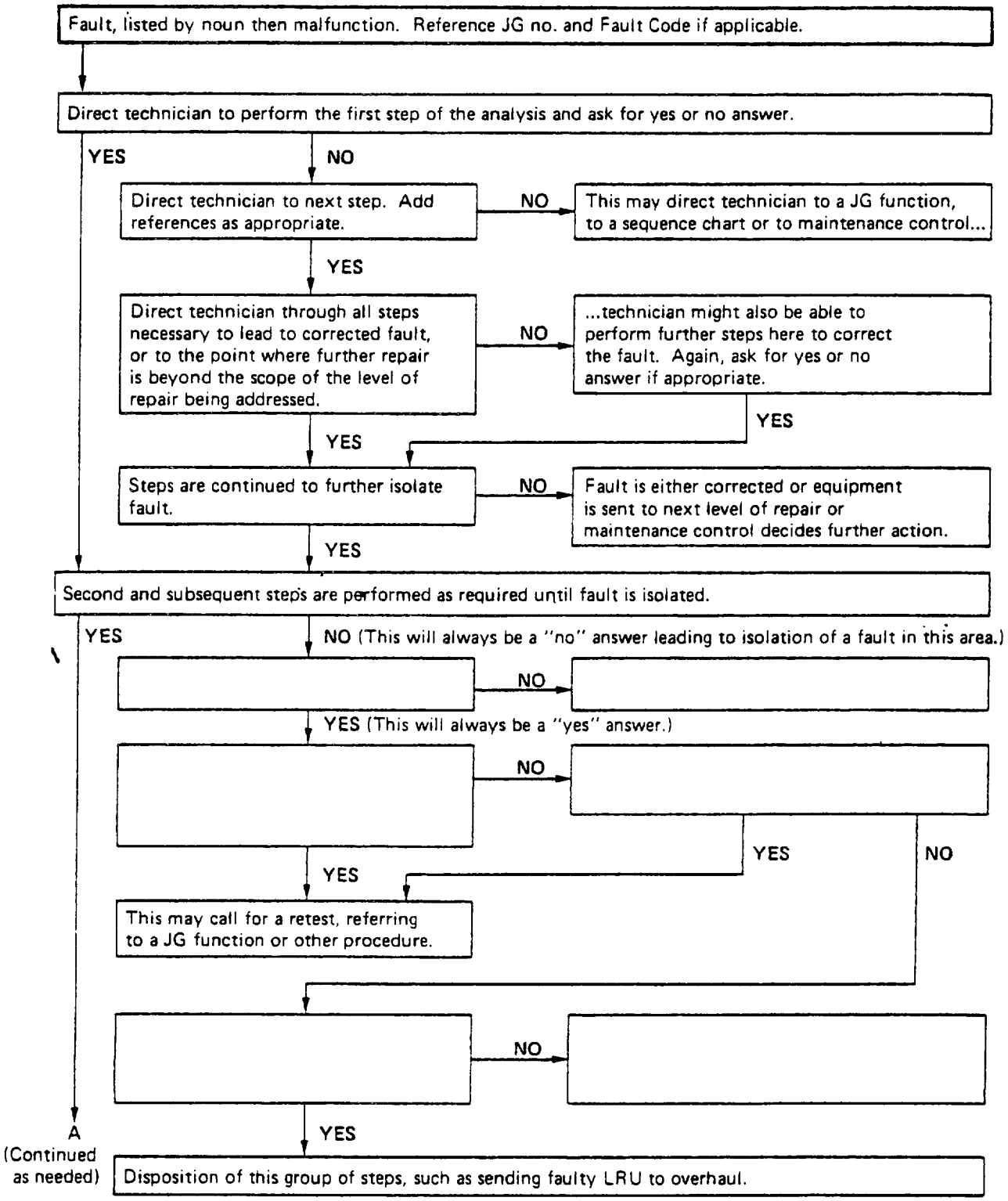


FIGURE 17. Troubleshooting diagram format guide.

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STEP	PROCEDURE	NORMAL INDICATION	ABNORMAL INDICATION	CORRECTIVE ACTION
1a	Open valve in 15 psig air supply line to emergency air conditioning unit EC-3 and test thermostat (27, table 2-2). Turn on a-c and d-c power to EC-3. Set circuit breaker CB-6 to the ON position and check industrial heater installed in paragraph 6-70, step 7.	Not heating	Heating	Check contactors K-9 and K-10. If either contactor is energized, replace energized contactor in accordance with table 6-1. If both contactors are energized, replace key switch KS-3 in accordance with table 6-1.
b	Set key switch KS-3 to the ON position. Check contactors K-9 and K-10.	Contactors K-9 and K-10 energize	Contactors K-9 and K-10 fail to energize	Refer to step 1, table 6-7.
c	Set test thermostat 10°F above thermostat ambient temperature and check industrial heater.	Industrial heater starts heating	Industrial heater fails to start heating	Refer to step 2, table 6-7.
2	Press manual starter MS-5 START button and check chilled water pump P-2.	Not operating	Operating	Adjust or replace pneumatic-electric switch PE-36 in accordance with paragraph 6-14.
3a	Set key switch KS-4 to the ON position. Close valve in 15 psig air supply. Bleed pressure from pneumatic lines in emergency air conditioning unit EC-3 by loosening flare fitting on pneumatic-electric switch PE-7A at test point 8 (figure 4-8). Tighten fitting after air pressure bleeds off. Check contactor K-11.	Contactor K-11 pulls in	Contactor K-11 fails to pull in	Refer to step 3, table 6-7.
b	Check chilled water pump P-2.	Pump starts	Pump fails to start	Refer to step 4, table 6-7.
c	Check voltage between test points 3 and 10 (figure 4-8).	28 (± 3) Vdc	Less than 25 Vdc	Replace contactor K-11 in accordance with table 6-1.

FIGURE 18. Tabular logic tree - checkout procedures and troubleshooting guide.

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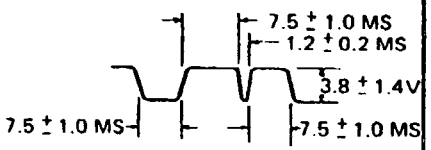
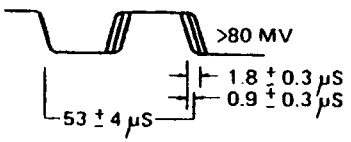
TEST OR PROCEDURE			INDICATION			INDICATION OBTAINED?	
LOCATION	ITEM	ACTION	LOCATION	INDICATOR	INDICATION	YES	NO
1. Transmitter Chassis	A1J1	Connect O-Scope	O-Scope	Scope Pushbutton Light	Lights	Go to 2	Replace A1 in Accordance with Para. 3-12
2. O-Scope	Scope Pushbutton	Press and Release	O-Scope	Channel B		Go to 3	Replace A1 in Accordance with Para. 3-12
3. Transmitter Chassis	A2J1	Connect O-Scope	....	....		....	....
4. O-Scope	a. Display Trig Source	A	....	....		....	....
	b. Time/Div Switch	10 μSec	O-Scope	Channel A		Go to 5	Replace A2 in Accordance with Para. 3-13

FIGURE 19. Tabular logic tree, troubleshooting chart.

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